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Key Points:
- Circular seismic surveys provide clear evidence for aligned fractures in the shallow critical zone (CZ), at depths of a few meters
- Shallow anisotropy aligns with fractures observed in boreholes, suggesting that inherited fractures persist through the weathering process
- Correlation between anisotropy and saprolite thickness suggests that inherited fractures provide access for meteoric water into the deep CZ

Supporting Information:
- Supporting Information S1

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Mapping Inherited Fractures in the Critical Zone Using Seismic Anisotropy From Circular Surveys

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Abstract Weathering and hydrological processes in Earth's shallow subsurface are influenced by inherited bedrock structures, such as bedding planes, faults, joints, and fractures. However, these structures are difficult to observe in soil-mantled landscapes. Steeply dipping structures with a dominant orientation are detectable by seismic anisotropy, with fast wave speeds along the strike of structures. We measured shallow (~2–4 m) seismic anisotropy using "circle shots," geophones deployed in a circle around a central shot point, in a weathered granite terrain in the Laramie Range of Wyoming. The inferred remnant fracture orientations agree with brittle fracture orientations measured at tens of meters depth in boreholes, demonstrating that bedrock fractures persist through the weathering process into the shallow critical zone. Seismic anisotropy positively correlates with saprolite thickness, suggesting that inherited bedrock fractures may control saprolite thickness by providing preferential pathways for corrosive meteoric waters to access the deep critical zone.

Plain Language Summary Bedrock undergoes physical and chemical changes near the Earth's surface that turns hard rock, such as granite, into softer, "weathered" rock. This soft rock acts as an important "sponge" that stores water used by vegetation and that feeds runoff in streams. Scientists are currently unable to predict how thick this sponge of soft rock will be at various locations. In this paper we test the idea that inherited cracks in the original bedrock provide pathways for water to percolate deeply into the subsurface, driving the weathering that turns hard rock into soft rock. We use a creative new design of circular seismometer arrays to detect these cracks, by measuring sound waves that travel more quickly along the cracks than across the cracks. Our results show that inherited cracks in the bedrock play an important role in the development of the soft-rock sponge that mantles Earth's surface.

1. Introduction

The complex uppermost portion of the Earth where rock, soil, water, air, and living organisms interact is referred to as the critical zone (CZ) (National Research Council, 2001). The CZ ranges from the top of vegetation down to fresh unaltered bedrock (e.g., Riebe et al., 2017) and controls many important processes, including river and ocean chemistry (Godsey et al., 2009), landscape evolution (R. S. Anderson et al., 2013), and the supply of nutrients to ecosystems (S. P. Anderson et al., 2002; Brantley & White, 2009; Pavich, 1986; Pavich et al., 1989). We define the CZ to encompass three main subsurface layers: soil, saprolite (highly chemically weathered rock), and a fractured/weathered bedrock layer (e.g., Befus et al., 2011; Holbrook et al., 2014). The properties and relative thicknesses of these layers, and their distribution across the landscape, are vitally important but difficult to measure and thus are poorly understood. CZ architecture is ultimately controlled by competition between the erosion of mobile regolith and the production of weathered materials at depth via physical, chemical, and biological processes (Brantley et al., 2007; Rempe & Dietrich, 2014). While surface erosion is relatively straightforward to measure (Dixon & Riebe, 2014), the complex physical and chemical processes that combine to determine the internal structure of the CZ are largely hidden from view. Casting light on those processes requires drilling and geophysics (e.g., Parsekian et al., 2015; Riebe et al., 2017).

Water plays a fundamental part in the complex weathering processes that create and maintain the CZ. Indeed, meteoric water chemically reacts with fresh bedrock until the water has reached chemical equilibrium (Brantley & White, 2009). Weathering processes are thus regulated by the hydraulic conductivity of the subsurface and the relative rates of groundwater flow and chemical reactions (Lebedeva &

Brantley, 2013; Maher, 2010; Maher & Druhan, 2014; Rempe & Dietrich, 2014). In addition, CZ properties may be influenced by environmental factors, such as frost cracking (R. S. Anderson et al., 2013) or climate (Rasmussen et al., 2011); physical factors, such as tectonic and topographic stresses (St. Clair et al., 2015); geological factors, such as lithology and legacy effects of climatic and tectonic history (Molnar et al., 2007); and human factors, such as agriculturally induced erosion (Mobley et al., 2015). A common theme across many of these ideas is the importance of high hydraulic conductivity flow paths that promote interaction of meteoric water with the surrounding rock (Riebe et al., 2017).

In crystalline bedrock terrains, hydraulic conductivity is mainly controlled by fractures (Gerke & Vangenuchten, 1993; Neuman, 2005; Zimmerman & Bodvarsson, 1996). In one study, the CZ was found to be thicker in strongly fractured crystalline rocks where fractures provide pathways for water to invade the bedrock, initiate weathering, and provide rapid bedrock drainage (Owen et al., 2007). Moreover, the flow of subsurface water is strongly influenced by parallel fractures (e.g., Shapiro et al., 2007), reinforcing the importance of understanding landscape-scale fracture patterns in crystalline bedrock (van Golf-Racht, 1982). The controls on water flow through saprolite are poorly understood, though some studies have found a zone of high hydraulic conductivity near the saprolite/bedrock boundary (e.g., Boisson et al., 2015). Given the definition of saprolite as weathered material that retains initial rock structure, bedrock structures (fractures, joints, and foliation) might persist well into the shallow subsurface and affect hydrologic flow. Characterizing subsurface fractures at landscape scales is difficult, however, especially when they are covered by soil and vegetation. Several geophysical methods have been used to study anisotropy due to subsurface fractures, including geophysical borehole logs (Hamm et al., 2007), azimuthal electrical resistivity soundings (Watson & Barker, 1999), self-potential gradient measurements (Wishart et al., 2008), and azimuthal seismic refraction surveys (e.g., Degnan et al., 2001; Hansen & Lane, 1995; Lu et al., 2009). However, each of these geophysical methods has disadvantages: drilling is expensive and invasive, and the resulting geophysical borehole logs only provide a 1-D representation; electrical soundings and self-potential gradients are largely limited to water-saturated fractures; and azimuthal seismic-refraction profiles are time-consuming, undersampled in azimuth space, and can conflate lateral heterogeneity with anisotropy.

Here we present an inexpensive and simple methodology to measure seismic anisotropy, and thus the dominant strike direction of subsurface remnant fractures within the saprolite, at the landscape scale. The approach is to acquire "circle shots" consisting of geophones deployed in a circle around a central shot point. Circular seismic surveys have been previously acquired at larger scales (e.g., to detect seismic anisotropy in oceanic crust and mantle; e.g., Stephen, 1985; Tsuji et al., 2011), but to our knowledge, such surveys have not been applied to address questions in CZ science. Seismic anisotropy is a measure of the directional dependence of wave speeds in a material. In geological materials, anisotropy can be caused by mineralogical fabrics (e.g., Godfrey et al., 2000), sedimentary layering (e.g., Wang, 2002), or oriented fractures (e.g., Crampin, 1984). On a circular array of geophones, if the subsurface velocity field is isotropic and homogeneous, and topography is planar, the compressional (P) wave will arrive simultaneously at all geophones. In the case of horizontal transverse isotropy (HTI) (e.g., aligned vertical fractures), the arrival times will produce a $\cos(2\theta)$ variation as a function of azimuth, with earlier arrival times along the fracture strike (Crampin, 1981). We applied this methodology to study near-surface remnant fractures in a weathered granite terrain in southeastern Wyoming (Figure 1). This site was chosen because it is well characterized by numerous boreholes (from surface to ~76 m depth) and seismic refraction surveys (Flinchum et al., 2017). While vertically oriented, isotropic laminar layering can also produce HTI anisotropy (e.g., Backus, 1965), this is ruled out by our field site being located in granite. Our results show that remnant fractures at 2–4 m depth in the shallow CZ saprolite are aligned with brittle, bedrock fractures at tens of meters depth in boreholes, indicating that fracture fabric endures throughout CZ weathering processes. Moreover, areas with greater seismic anisotropy—and thus presumably higher fracture density—have thicker saprolite, suggesting that inherited bedrock fractures may influence weathering processes and fluid flow in the shallow CZ.

2. Site Description and Geophysical Surveys

2.1. Study Site

The Blair-Wallis (BW) catchment is located in a Precambrian (1.43 Ga) weathered granite terrain in the Laramie Range, in southeastern Wyoming (Figure 1). The site is composed of coarse-grained, biotite hornblende

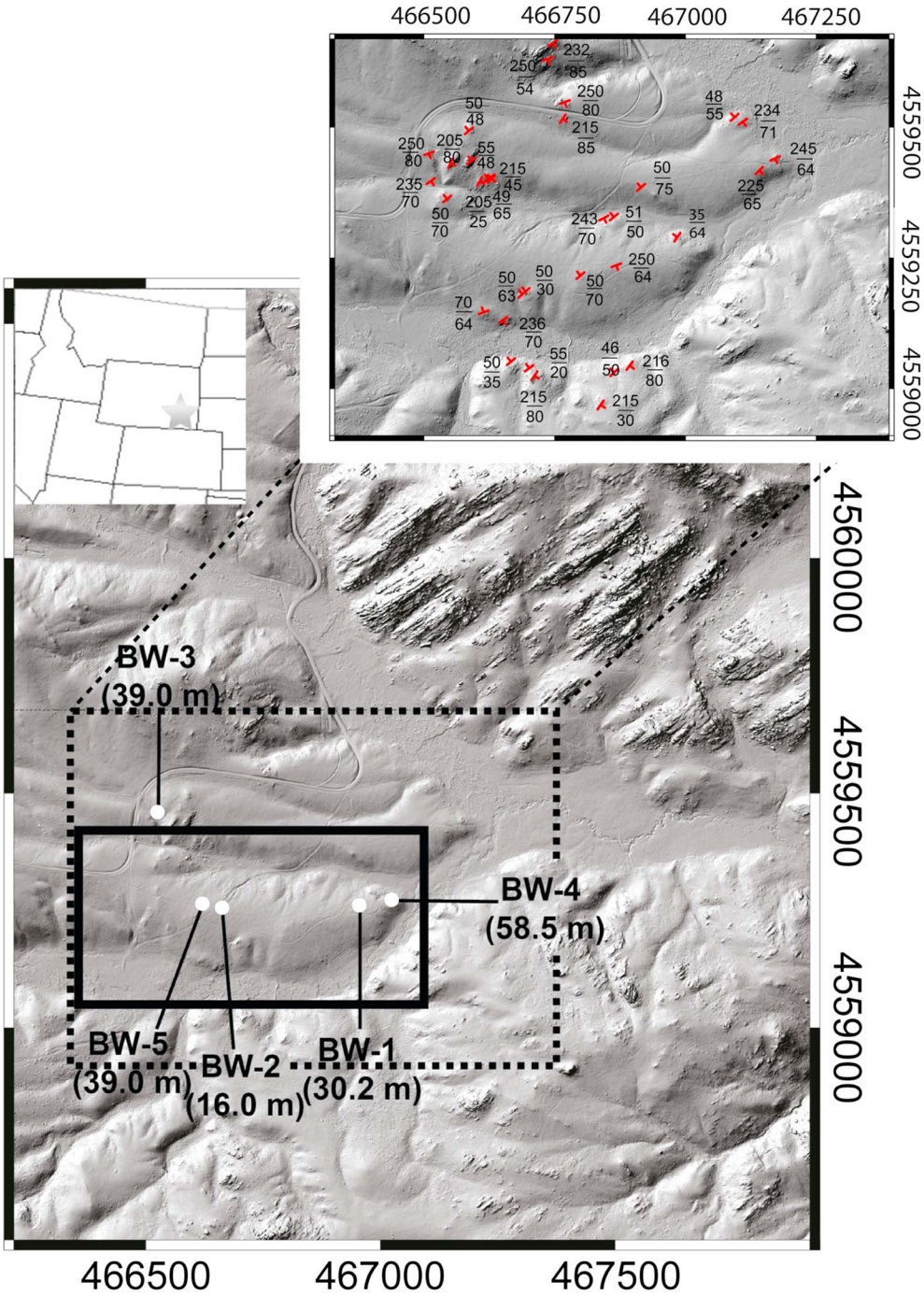

Figure 1. Location of study site in southeastern Wyoming, showing shaded topography from LiDAR data. Bold box shows ridge at BW site, along with locations of five boreholes (Figure 2a); numbers beneath borehole names denote total depth of hole. Prominent topographic fabric in northeastern part of the map is due to NE-SW fractures in granitic bedrock. Inset at upper right shows strike/dip measurements on outcrops at the ridge (dotted box in main map), which show the same dominant trend. Axes are in UTM coordinates (meters), referenced to UTM grid 13°N.

Sherman granite (Frost et al., 1999), whose fabric exhibits no apparent lattice preferred orientation. The BW area is characterized by a smooth, undulating erosional surface with prominent granite tors and outcrops rising out of thin soils in sections of the landscape (Eggler et al., 1969). In aerial photographs, Sherman granite outcrops show a dominant NE-SW fracture direction (Figure 1). The site has a mean annual temperature of 5.4°C and receives 620 mm of annual precipitation, of which 90% is in the form of snow

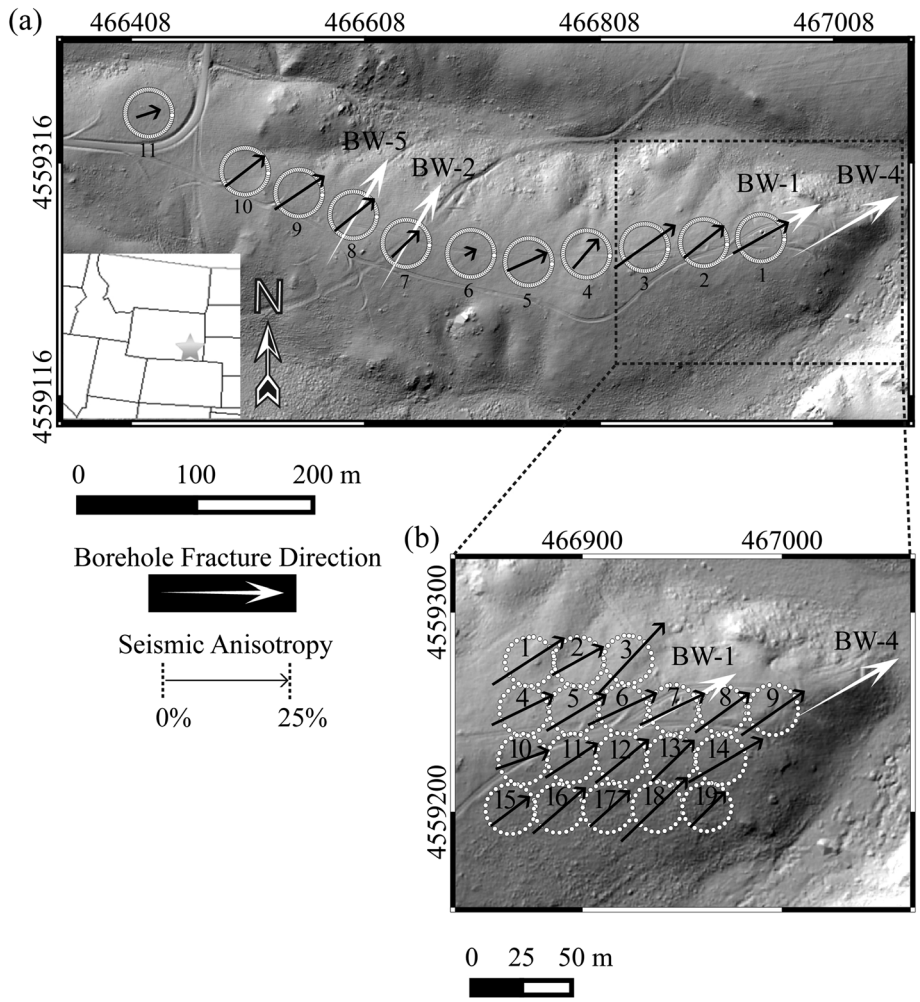

Figure 2. Locations of circle shots collected at BW for the (a) ridge line and (b) hilltop surveys. Geophone locations are marked by white dots. The compass orientation of the dominant anisotropy direction is marked by black lines. Line length corresponds to the magnitude of anisotropy. The dominant strike directions of bedrock fractures measured at >20 m depth in the boreholes are marked by white arrows.

(Natural Resources Conservation Service, 2015). Flora at the site comprise grasslands and sagebrush with patches of Aspen groves at the foot of the catchment (United States Department of Agriculture, 2005). Five boreholes are located in the study site with final depths ranging from 16.5 m to 60.4 m (Figure 1). We measured fracture orientations (strike and dip) on surface outcrops using a Brunton compass and in boreholes using acoustic (ATV) and optical televiewer (OTV) data (Figure 2; supporting information Figure S1). Subsurface velocity structure across the catchment was determined from a 3-D seismic refraction volume created from over 30 seismic refraction profiles (supporting information section S2 and Figures S2–S5; Flinchum et al., 2017). Comparison of drilling and refraction results shows that the saprolite-weathered bedrock boundary occurs at the 1,200 m s^{-1} contour (Figure S5).

2.2. Geophysical Surveys

Two different circle-shot surveys were conducted in July 2016, with the first consisting of 19 circle shots with circle radii of 12.5 m, and the second with 11 circle shots of radius 20 m (Figure 2; Novitsky & Holbrook, 2017). The two circle sizes provided measurements at two different depths of penetration and enabled spatial coverage at different scales. The 20 m circle shots were deployed along an entire ridgeline to study anisotropy at landscape scales (~600 m or "ridgeline" scale). The 12.5 m circles were deployed in a tight grid to characterize the consistency of anisotropy at a smaller scale (~100 m or "hilltop" scale). For both surveys, 10 Hz, vertical-component geophones were evenly distributed along a fixed-radius circle centered on a source point (Figure 2). The

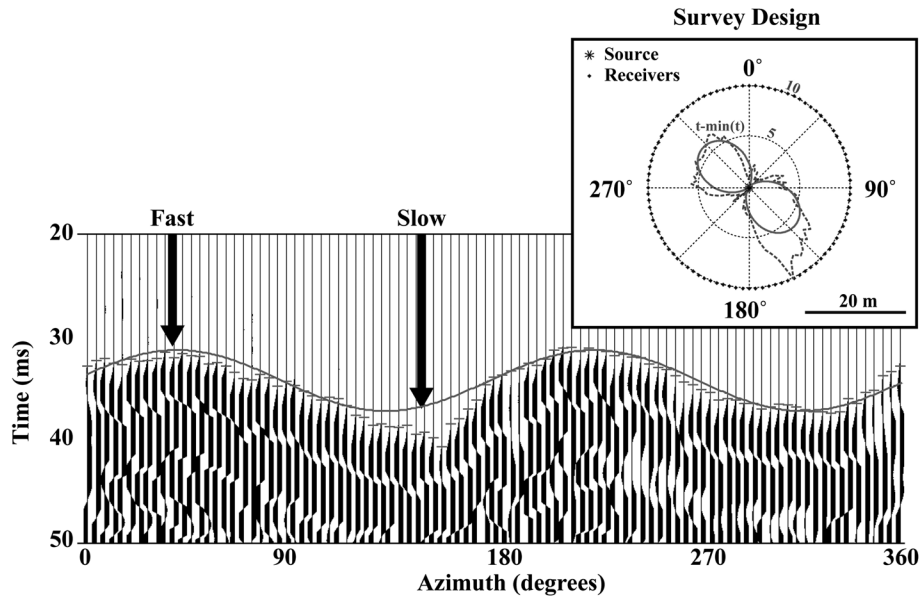

Figure 3. Seismic data from a circle shot with 20 m radius centered on the BW-5 well location (circle shot #8). The *x* axis is azimuth measured clockwise from north of each geophone around the circumference of the circle. Manually picked first arrivals are marked by short horizontal lines. Zones of early (fast) and late (slow) first arrivals are spaced approximately 90° apart, indicating azimuthal anisotropy in subsurface seismic velocities. Solid line shows predicted travel times from model. Inset shows survey design and superimposed Rose diagram plot of travel time differences from the minimum arrival time (dashed = observed, solid = predicted), showing clear pattern of azimuthal anisotropy. The "5" and "10" are isochrons in milliseconds.

seismic source consisted of a 4.5 kg sledgehammer striking an aluminum plate. At each shot location, seven individual hammer swings were stacked to increase signal-to-noise ratio. For the ridgeline survey, we deployed 96 geophones at equal spacing at 20 m radius to record 11 circle shots spaced at 50 m along the ridge. The hilltop survey consisted of 19 circle shots with 24 geophones arrayed in 12.5 m radius circles in a dense grid around the BW-1 borehole on the ridge crest. For each circle shot, the source and receiver locations were GPS surveyed, and the topography was extracted from airborne LiDAR data available in the area. *P* wave first arrival times were manually picked on all traces with sufficiently high signal-to-noise ratios and represented at their corresponding azimuths, calculated from GPS measurements and corrected to true north.

On all circle shots acquired in our site, the dominant arrival pattern shows a sinusoidal variation of first arrival times, with two slow (late) directions and two fast (early) directions (Figure 3; supporting information section S3 and Figures S6 and S7). We modeled this as an HTI medium, which produces a $\cos(2\theta)$ variation in travel time as a function of azimuth, where seismic waves propagate faster along remnant fracture strike than perpendicular to it. First arrivals were fitted with a least squares $\cos(2\theta)$ inversion with fast and slow directions determined by the shortest and longest travel times, respectively. The percentage of seismic anisotropy was then calculated using the equation by Birch (1960):

$$\%\text{Anisotropy} = \frac{V_f - V_s}{\frac{(V_f+V_s)}{2}} \times 100 \cong \frac{T_f - T_s}{\frac{(T_f+T_s)}{2}} \times 100 \tag{1}$$

where V_f is the fastest velocity, V_s the slowest velocity, T_f the shortest first arrival, and T_s the longest first arrival. We assumed that all raypaths around the circle have identical lengths so that we can substitute travel time for velocity in equation (1).

The error in the estimated percentage of anisotropy was calculated via bootstrapping (Sacchi, 1998), which randomly removes a predetermined fraction of the data set prior to inversion. The error is then estimated from the standard deviation of the set of inversion results (e.g., Parsekian & Grombacher, 2015). We randomly removed 50% of the original data and fit the subsampled data with a least squares $\cos(2\theta)$ curve. We repeated this operation 400 times for each circle shot. We finally estimated the error in the anisotropy from the standard deviation of all the inversions, following Parsekian and Grombacher (2015).

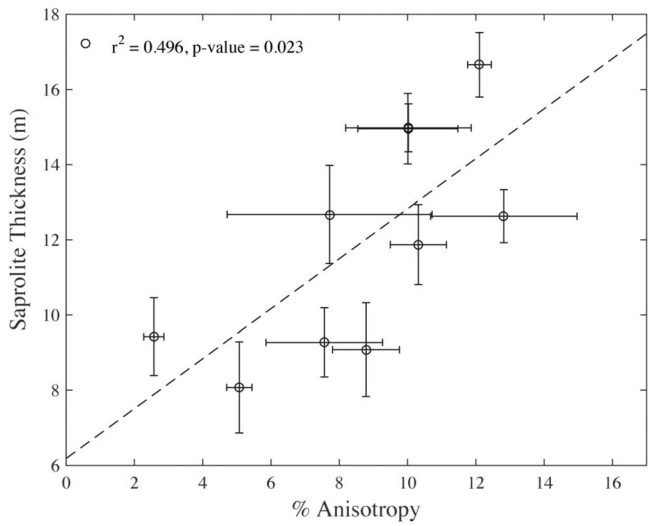

Figure 4. Crossplot of the percentage of seismic anisotropy versus regolith thickness. Seismic anisotropy was estimated from the circle shots, while regolith thickness was extracted at the location of each circle shot from a 3-D volume of seismic refraction velocities interpolated from inversions of numerous 2-D seismic refraction profiles (Figure S2; Flinchum et al., 2017).

3. Results

The inversion produces an estimate of the magnitude and direction of anisotropy at each circle-shot location (Figure 2). The results indicate a clear dominant anisotropic fast direction oriented NE-SW at both the ridgeline and hilltop scales. The fast direction of anisotropy matches the strike direction of fractures visible on LiDAR topography (Figure 1), hand-measured on outcrops (Figure 1), and determined from borehole fractures picked on oriented ATV and OTV images between 15 and 60 m depth (Figure 2; cf. supporting information Figure S1). The strike of fractures observed in holes BW-5 and BW-2 is ~N 030°; the fast direction of the colocated circle shots ranges between N 015° and 020°. At the location of borehole BW-1, the fracture strike direction inferred from both the borehole images and the circle shots is ~N 060°.

For each ridgeline circle shot, we extracted the depth of the base of saprolite from a 3-D volume of seismic refraction velocities (Flinchum et al., 2017; Figures S2–S5). The error in saprolite depth was estimated as the standard deviation of 360 depth estimates, located on a circle 10 m from the center of each circle shot; 10 m was chosen because it represents the source/receiver midpoint for the 20 m radius circle shots. A linear regression between seismic anisotropy and saprolite thickness shows a positive correlation (r^2 = 0.496 and p = 0.023; Figure 4). (We omitted circle shot #3 (Figure 2a) because a large quartz vein observed there indicates a nongranite lithology.)

4. Discussion

Our results demonstrate an approach to characterizing oriented structures in the shallow CZ. While the circle-shot geometry has been applied in at least one study to characterize small-scale fracture anisotropy in an outcrop (Vilhelm et al., 2010), we are unaware of any prior studies that use this approach to study CZ processes, nor at the landscape scale. Any structures that might create azimuthal anisotropy are amenable to study with this approach. Here we assess possible caveats in the method and then speculate about potential implications for CZ development.

We consider five caveats to our approach: nonplanar topography, lateral velocity heterogeneities, dip of fracture planes, multiple fracture sets, and the need for 2-D refraction data. First, to mitigate the effects of uneven topography, we endeavored to place all circle shots on nearly planar surfaces so first-arrival times would not be influenced by topographic undulations (supporting information section S4 and Figures S8 and S9). No correlation is observed between geophone elevation and arrival time. We note that our method does not assume that topography is level but merely planar, including planar hillslopes (Figure S10). Deviations from planar topography will cause variability in arrival times that may be difficult to correct without full raytrace modeling. Second, lateral velocity heterogeneities or laminar layering should be minimal in the relatively uniform granite lithology at our site (with the exception of the quartz vein mentioned above). This expectation is verified by the remarkable consistency of the inferred anisotropy directions at the hilltop scale (Figure 2b). Third, our method will only detect azimuthal anisotropy, such as that produced by an HTI medium with steeply dipping fractures (i.e., near vertical). If fractures are all horizontal, for example, no anisotropy will be observed on the circle geometry. To quantify this effect, we modeled the effect of fracture dip on the magnitude of observed seismic anisotropy for a tilted transverse isotropy (TTI) medium using an anisotropic elastic wave propagation code (Okaya & McEvilly, 2003). The results indicate that fracture dip must exceed 45° to produce significant travel time variations in the circle shots (supporting information Figures S5, S11, and S12). This condition is largely met at our site, since surface outcrops predominantly show steeply dipping fractures (Figure 1) and borehole measurements show many steeply dipping fractures (supporting information section S1). Fourth, the circle-shot method works most straightforwardly if there is a single dominant fracture strike, since multiple fracture azimuths would produce travel times that do not fit the cos(2θ) pattern. (Multiple-azimuth patterns might be interpretable, however.) Finally, acquisition of circle shots should be accompanied by standard,

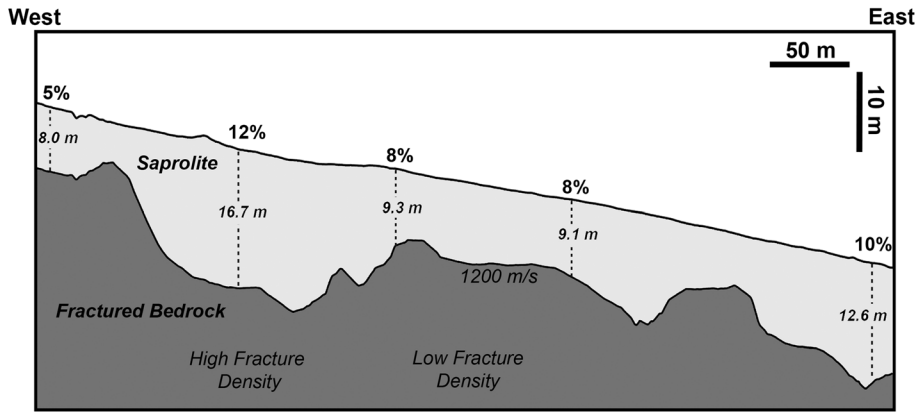

Figure 5. Cross section along the crest of the ridge showing seismic anisotropy and regolith thickness. The profile runs east-west from circle #11 to circle #1. Regolith thickness was extracted from a 3-D volume of seismic refraction velocities, assuming that the 1,200 m/s velocity represents the base of saprolite (Figure S5). Surface topography was extracted from the LiDAR data along the profile.

multioffset seismic refraction data. Circle shots alone provide no information about the distribution of seismic velocity at depth or the penetration depth of the seismic energy.

Despite these caveats, the circle-shot method can quickly and accurately determine remnant fracture azimuths in the shallow subsurface CZ. Results at both the hilltop and ridgeline scales show coherent signals consistent with a NE-SW (~N 055°) fracture strike direction (Figure 2).

The agreement of fracture orientations in the shallow subsurface with those observed at greater depths in the boreholes indicates that even highly weathered saprolite retains structural characteristics of the deep parent rock. While this is not unexpected and indeed conforms to the standard definition of saprolite (Merrill, 1897), our results provide direct evidence that fractures persist through the weathering process and create measurable physical property contrasts within saprolite. This raises the possibility that inherited structures in saprolite could influence CZ development. Inherited fractures and joints, especially if vertically oriented, could facilitate the infiltration of meteoric water into bedrock and enhance the downward propagation of weathering (Aarão Reis & Brantley, 2017). We might then expect an irregular weathering front at the base of saprolite, depending on spatial heterogeneity in inherited fracture density. This hypothesis would predict a correlation between seismic anisotropy and saprolite thickness, since higher fracture density should produce both greater anisotropy and (potentially) thicker saprolite due to locally higher hydraulic conductivity. Indeed, such a correlation is observed at our site (Figure 4). A synthesis cross section along the ridgeline shows that both saprolite thickness and anisotropy are highly variable and correlated (Figure 5). We suggest that bedrock preferentially weathers in highly fractured areas, resulting in the undulating thickness of regolith under the ridgeline. We are unaware of any other existing model of CZ development that predicts such strong heterogeneity in saprolite thickness along a ridge crest.

Incorporating the notion of inherited fractures could improve our understanding of CZ processes and help understand or locate areas of variable saprolite thickness. Existing models of CZ development assume homogeneous bedrock and do not incorporate inherited fractures. As rock is exhumed toward the surface, fractures should open under the relaxing stress field (St. Clair et al., 2015); zones of denser inherited fractures should thus open up preferentially, enhancing water access into the CZ and causing greater weathering. Irregular saprolite thickness may thus result from changes in chemical or physical (dip or fracture density) properties of the bedrock. In relatively homogeneous bedrock (such as the Sherman granite studied here), inherited fractures could be a significant factor influencing the development of the CZ.

Finally, we offer two points for future applications of circle-shot seismic anisotropy studies of inherited fractures in the shallow CZ. First, a current trend in portable active-source seismology is the use of very large numbers of small easy-to-deploy recording instruments. A novel future experiment could use a dense 2-D array of these instruments draped across the area of interest. Any number of sources can be collected by this 2-D array. Seismograms for circle-shot analysis can subsequently be extracted during postfield data processing. This approach is more efficient during field acquisition and will result in a larger volume of data; for

example, circle shots of increasing radii will provide information at increasing depths (e.g., examination of two-layer anisotropy). Internal instrument spacing within the 2-D array must be strategically selected in order to minimize nonuniform radial distances upon circle extraction. Second, we note that a visible asymmetry in the sinusoidal pattern of arrival times exists at selected circle-shot locations (Figure 3 and supporting information section S3). Conceptually, this second-order effect can indicate the presence of two-layer anisotropy, dominant/subordinate conjugate fracture sets, or, in stratified sediments, a regional fracture field superimposed onto tilted sediments. These second-order asymmetric arrival times require more advanced anisotropy analysis methods but would potentially offer additional information about CZ structure and geometry of multiple fracture sets.

5. Conclusions

We demonstrated an inexpensive and simple geophysical methodology to detect oriented fractures in the subsurface, determine their dominant strike direction, and qualitatively evaluate fracture density. The approach uses circle shots to determine the direction and magnitude of seismic anisotropy in the upper 2–4 m, which can be directly related to subsurface inherited fractures in the saprolite. We successfully applied this methodology in a weathered granite terrain, at both ridgeline and hilltop scales. Our results showed good consistency between the main fracture directions estimated from circle shots at both scales and those observed on surface outcrops and measured in nearby geophysical boreholes that measure in situ bedrock fractures. The results also highlighted a clear correlation between seismic anisotropy estimated from circle shots and saprolite thickness determined from a 3-D seismic refraction volume. Our results suggest that the inherited fracture characteristics of bedrock may control the thickness of regolith at our site and that the regolith itself retains the original fracture fabric of the parent bedrock—consistent with previous definitions of saprolite (Dixon et al., 2009; Riebe & Granger, 2013). This methodology has the potential to bring greater understanding to hydrology and surface processes in the shallow CZ at a landscape scale.

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